Unit 1

Lesson 2

# Spark The Challenge!

## Smart Starts.

**Time Frame: [90 min]**

### Lesson Objectives:

**By the end of this lesson students will be able to:**

- Identify a problem that makes it hard for someone with a disability to use a space in the home.
- Think of possible ways to help people who use a wheelchair move around more easily in their homes.
- Describe the job of an Assistive Technology Specialist in helping others choose the right tools.
- Recognize how kindness and problem-solving are part of creating helpful technology for all.

## Challenge questions of the lesson

1. How can we help someone who uses a wheelchair move easily around their home?
2. How can we work together to share our ideas about making a room easier to use?
3. How can we draw or build a model of a room that everyone can use?

## SEL Relation

- **Self-Awareness:** Students consider how accessibility challenges affect Uncle Sam, building empathy by imagining his experience.
- **Social Awareness:** Through stories and discussion, students learn to respect differences and understand challenges faced by people with disabilities.
- **Relationship Skills:** Group activities encourage clear communication, active listening, and teamwork to solve problems together.
- **Responsible Decision-Making:** Using the Engineering Design Process, students identify problems, brainstorm, and choose solutions that meet others' needs.
- **Self-Management:** Students practice perseverance and adaptability as they design, test, and improve their models.

# Engage

## Teacher's Role:

- Facilitate a welcoming and inclusive discussion to create a safe space for all students.
- Introduce the story scenario about Uncle Sam, who uses a wheelchair and wants to move easily around his home.
- Use storytelling with props to bring the scenario to life, supporting comprehension through visuals and gestures.
- Encourage participation from all students by inviting them to listen actively and share ideas verbally, through drawing, or acting.
- Use clear, simple language and support understanding with pictures or gestures, especially for language learners or students with special needs.

## Expected From Students:

- Listen attentively to the story of Uncle Sam and his accessibility hallenges.
- Engage by sharing ideas about the story through talking, drawing, or acting.
- Participate in discussions prompted by questions about accessibility and inclusion

## Parts of the books included.


The next day, Laila talks to Miss Sara:

Laila: "Miss Sara, I want to help my Uncle Sam turn on his lamp by using his voice. But I don't know how. Can you help me?"

Miss Sara (smiling): "That's a great idea, Laila! But before we think about the lamp, we need to make sure his home is easy for him to use. That's called accessibility."

Laila (curious): "What does accessibility mean?"

Miss Sara: "It means making sure your uncle can move around and use things in his home without trouble. We can look at where the furniture is, how high the shelves are, and if he can reach things from his wheelchair."

Laila: "Oh! So we need to make his home more comfortable for him first?"

Miss Sara: "Yes, exactly! After that, we can use smart tools like voice-controlled lights to help even more."

Laila (excited): "That's a good plan! I'll start thinking about how to fix up his home today!"

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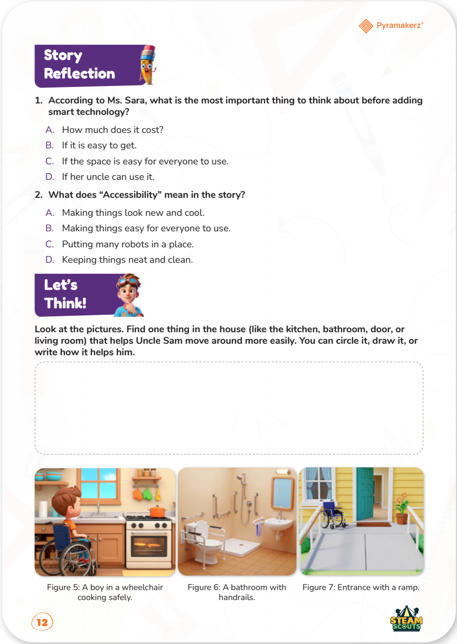
Pyramakerz®

Story Reflection

1. According to Ms. Sara, what is the most important thing to think about before adding smart technology?
   A. How much does it cost?
   B. If it is easy to get.
   C. If the space is easy for everyone to use.
   D. If her uncle can use it.
2. What does "Accessibility" mean in the story?
   A. Making things look new and cool.
   B. Making things easy for everyone to use.
   C. Putting many robots in a place.
   D. Keeping things neat and clean.

Let's Think!

Look at the pictures. Find one thing in the house (like the kitchen, bathroom, door, or living room) that helps Uncle Sam move around more easily. You can circle it, draw it, or write how it helps him.

Figure 5: A boy in a wheelchair cooking safely.

Figure 6: A bathroom with handrails.

Figure 7: Entrance with a ramp.

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# Story

### Set the Scene with Curiosity and Care

Teacher says (gently, with interest):

- Read or act out the scene where Lila meets Ms. Sara and expresses her desire to help Uncle Sam control his lamp with his voice.
- Emphasize Ms. Sara's explanation that accessibility means designing a space that allows Uncle Sam to move and use everything easily, before adding smart technology.
- Highlight Lila's questions and growing understanding to model curiosity and thoughtful problem-solving.

As the teacher, you've just read the scene where Lila seeks Ms. Sara's help for Uncle Sam. Design three different strategies to build upon this scene to deepen student understanding of accessibility and inspire creative solutions. Consider these factors:

## Possible Strategies for Implementation:

- **Interactive Engagement**

  Use role-playing, storytelling with props, or visual displays to bring the characters and challenges to life. Allow students to act out scenes, use simple costumes, or manipulate objects to deepen their understanding and boost participation.

- **Inclusive Differentiation**

  Provide options to support various learning styles and needs—such as illustrated storyboards for visual learners, hands-on building for kinesthetic learners, and sentence starters or word banks for language learners. Pair students for peer support and use sentence frames or visuals as scaffolds.

- **Meaningful Real-World Connections**

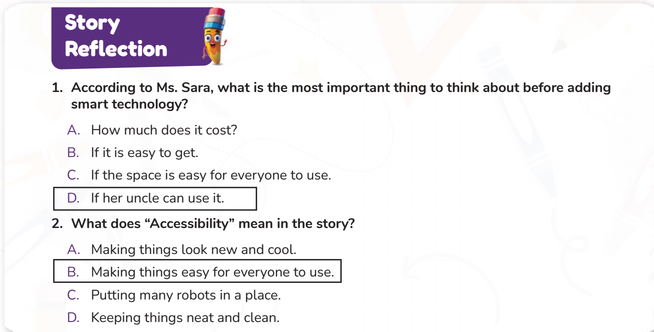
  Relate the story's themes (e.g., empathy, accessibility, smart tools) to everyday experiences. Invite students to share stories from their families or communities, or guide them in identifying accessibility features they've seen (like ramps, elevators, or voice assistants at home or school).

# Story Reflection

### Objective:

Reflect on key concepts from the story by answering comprehension questions focused on accessibility and smart technology.

Answer: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## Possible Strategies for implementation:

1. Think-Pair-Share
   - How to Use:
     - Think: Give students 1–2 minutes to think about each question and jot down their answers.
     - Pair: Have students pair up and discuss their answers and reasoning.
     - Share: Invite pairs to share their thoughts with the class. This encourages participation and helps clarify misconceptions.
2. Four Corners
   - **How to Use:**
     - Assign each answer choice to a corner of the room.
     - Read the question aloud. Students move to the corner representing their answer.
     - In each corner, students discuss why they chose that answer.
     - Ask a representative from each group to explain their reasoning to the class.
     - Debrief and reveal the correct answer, discussing why it's correct.
3. Role Play
   - **How to Use:**
     - Assign roles (e.g., Ms. Sara, Lila, Uncle Sam) to students.
     - Have them act out a scene where they discuss adding smart technology and what accessibility means.
     - After the role play, discuss as a class which answer choices best match the ideas in the scene.

## Let's Think!

**Objective:**

Students will explore how to make a home easier for someone using a wheelchair by circling, drawing, or writing about helpful features—promoting observation, communication, and empathy through design thinking.

Answer:

**Possible correct answers from the pictures:**

- **Kitchen:**
  - The lowered countertop and open space under the sink/stove allow Uncle Sam to roll his wheelchair close and use the kitchen independently.
- **Bathroom:**
  - Grab bars near the toilet and shower help Uncle Sam transfer safely.
  - The roll-in shower (no step) makes it easy for a wheelchair to enter.
  - The sink is at a height accessible from a wheelchair.
- **Door/Entrance:**
  - The ramp instead of stairs allows Uncle Sam to enter and exit the house safely and independently.
  - The wide doorway accommodates a wheelchair.

**Students might circle, draw, or write about any of these features.**

## Possible Strategies for implementation:

1. **Think-Pair-Share**

- **Think:** Give students time to look at the pictures and think about what helps Uncle Sam.
- **Pair:** Have students discuss their ideas with a partner.
- **Share:** Invite pairs to share their answers with the class.

2. **Visual Marking**

- Students can
  - Circle the accessibility feature directly on the picture.
  - Draw an arrow or sketch the item (e.g., ramp, grab bar).
  - Write a sentence explaining how it helps (e.g., "The ramp lets Uncle Sam get inside.").

3. **Group Discussion**

- Display the pictures on the board.
- Ask students to come up and circle or point out features.
- Discuss as a class why each feature is helpful.

4. **Role Play**

- Have a student pretend to be Uncle Sam and describe how they would use each feature (e.g., "I can roll up the ramp to get inside!").

# Explorer's Toolkit

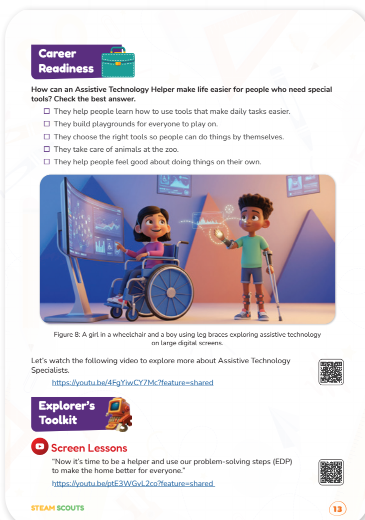
## Teacher's Role:

- Begin by showing the image on the right and ask students to describe what they see. Encourage them to look closely at the symbols (keyboard, speaker, screen reader, etc.).
- Guide a "See, Think, Wonder" routine:
    1. See: What do you notice in this image?
    2. Think: What do you think these tools are used for?
    3. Wonder: What do you wonder about how these tools help people?
- Link their responses to the job of an Assistive Technology Helper.
- Read the question aloud and go through the answer options one at a time. Ask students to raise their hands if they think each one helps describe what this career does.
- Lead a discussion on how different tools help people with different needs, and why choosing the right one matters.
- Encourage students to work in pairs to explain their chosen answers using simple examples from daily life.

## Expected From Students:

- Share their observations and thoughts using the "See, Think, Wonder" prompts.
- Discuss with a partner which answer best matches the role of an Assistive Technology Helper.
- Reflect on how assistive tools like screen readers, ramps, or talking devices help people be more independent.
- Select and explain the best answer(s) from the list.

## Parts of the books included.v


Career Readiness

**How can an Assistive Technology Helper make life easier for people who need special tools? Check the best answer.**

- ☐ They help people learn how to use tools that make daily tasks easier.
- ☐ They build playgrounds for everyone to play on.
- ☐ They choose the right tools so people can do things by themselves.
- ☐ They take care of animals at the zoo.
- ☐ They help people feel good about doing things on their own.

Figure 8: A girl in a wheelchair and a boy using leg braces exploring assistive technology on large digital screens.

Let's watch the following video to explore more about Assistive Technology Specialists.

https://youtu.be/4FgYiwCY7Mc?feature=shared

Explorer's Toolkit

Screen Lessons

"Now it's time to be a helper and use our problem-solving steps (EDP) to make the home better for everyone."

https://youtu.be/ptE3WGvL2co?feature=shared

STEAM SCOUTS 13

## Career Readiness

**Objective:**

Students will understand the role of an Assistive Technology Helper by identifying how this career supports people with special needs through the use of tools that promote independence and accessibility in daily life.

**Video Link:** https://youtu.be/4FqYiwCY7Mc?feature=shared

**Possible Best Answers:**

- They help people learn how to use tools that make daily tasks easier.
- They choose the right tools so people can do things by themselves.
- They help people feel good about doing things on their own.

These three answers best represent the role of an Assistive Technology Helper.

## Strategies for implementation:

1. **Pre-Viewing Discussion**

**Strategy:** Activate Prior Knowledge

- Implementation: Ask students, "Have you ever used something that helped you do a task more easily?" (e.g., a pencil grip, big scissors, or a buttoning tool).
- This helps students connect the role of assistive tools to their own experience before learning about Assistive Technology Specialists.

2. **Guided Video Viewing**

**Strategy:** Focused Listening with a Purpose

- Implementation: Before playing the video, give students a focus question: "What kinds of tools do Assistive Technology Helpers use to make life easier?"
- Pause the video after each example and invite quick reflections through partner talk.

3. **Interactive Discussion with Visuals**

**Strategy:** Use of Real-Life Visual Aids or Objects

- Implementation: Show pictures or real items (e.g., grabbers, large-print books, voice activated devices).
- Let students match them with actions (e.g., "This helps someone read" or "This helps someone grab something from a shelf").

4. **Think-Pair-Share**

**Strategy:** Collaborative Thinking

- Implementation: After watching the video, students choose 2 checkboxes they think are most important and discuss their reasoning with a partner before sharing with the class.

### 5. Career Role Play (Optional Extension)

**Strategy:** Role Simulation

- Implementation: Students act as “Assistive Technology Helpers” helping classmates solve small challenges using pretend or simple adaptive tools.
- Helps reinforce understanding of the specialist’s role while building empathy and problem-solving.

## Explain

### Teacher Role:

- Set the scene by saying: “Today, we’ll think like real engineers! We’ll see how we can make a playground that everyone can enjoy—no matter their needs.”
- Preview the term EDP and list the steps: **Ask – Imagine – Plan – Create – Test – Improve.**
- Play the video and pause at each step to ask students what it means and how it helps solve problems.
- Encourage students to relate the steps to real playground issues (e.g., ramps instead of stairs, sensory paths, wide walkways).

### Expected From Students:

- Watch the video and recognize the six EDP steps in action.
- Participate in guided pauses to reflect on each step.
- Apply what they learned to brainstorm playground improvements for accessibility.
- Begin sketching or describing their own inclusive design idea using the EDP model.

### Parts of the books included.

**Screen Lessons**

“Now it’s time to be a helper and use our problem-solving steps (EDP) to make the home better for everyone.”

https://youtu.be/ptE3WGvL2co?feature=shared

STEAM SCOUTS 13

## Screen Lessons

**Objective:**

Students will understand how to use the Engineering Design Process (EDP) to create a more inclusive playground that supports all children, including those with disabilities.

**Video Link:** https://youtu.be/ptE3WGvL2co?feature=shared

## Possible Strategies for implementation:

1. **Pre-Watching: Set a Purpose**
   - **Strategy:** Prediction Prompt
   - **How to Implement:**
     1. Tell students: “We’re about to meet a team using something called the Engineering Design Process—EDP—to fix a problem. What do you think the problem is?”
     2. Write or draw predictions on the board. Say: “Watch closely and see if your guess is right.”
2. **During Watching: Active Engagement**
   - **Strategy:** EDP Step Tracker
   - **How to Implement:**
     1. Give students a printed or drawn chart with the 6 steps of EDP: Ask – Imagine – Plan – Create – Test – Improve.
     2. As they watch, students check off each step when they notice it happening in the video.
     3. Encourage drawing a symbol or writing one word next to each step.
3. **Post-Watching: Discussion and Reflection**
   - **Strategy:** Think–Pair–Share + Real-World Link
   - **How to Implement:**
     1. Ask: “What did the team do that made the playground better for everyone?”
     2. Let students first think quietly, then talk to a partner, then share with the whole class.
     3. Guide them to connect the team’s actions to things they could improve in their own school playground using the EDP steps.

# Engineering Design Process

Name: ..............................................................

Project: ..............................................................

## Ask

**What is the problem?**

- Some people, like Uncle Sam, have trouble moving, seeing, or hearing.
- It's hard for them to use regular lights, doors, or furniture at home.

## Imagine

**What ideas can help?**

- Use a smart light that turns on with voice commands.
- Make wide paths and low shelves for easier movement.
- Add grab bars near beds or in the bathroom.
- Use PictoBlox to control a sprite or LED by saying "Lights on."

## Plan

**What will we build?**

- A small model of a smart room (bedroom, kitchen, or bathroom).
- The room will have one voice-controlled object (e.g., lamp or fan).
- It will include design features for someone who uses a wheelchair or has hearing/vision challenges.

## Create

## Improve

## Now I Can

### Objective:

To help students reflect on their learning and recognize how their actions and thinking have supported inclusion, problem-solving, and empathy through STEAM and design.

### Activity Instructions

#### Step 1: Create a Calm, Supportive Reflection Moment

- Begin with:

  "Let's take a few minutes to think about everything we learned and how we became smart home designers who care about everyone!"

- Set a quiet and positive mood.

#### Step 2: Introduce the "Now I Can" Checklist

- Show the checklist on the board or projector and read each line slowly.
- After each one, pause to ask:

  "Give me a thumbs up if you can do this now!"

  "Can anyone share something they did to help someone in their smart home model?"

- This allows for oral reflection and supports different learning styles.

#### Step 3: Reflect with Drawing or Writing

- Pass out the checklist with small boxes for students to check or color stars.

  Ask:

  "What is one part of your model, or one idea you shared, that could help someone with a disability?"

- Let them draw it or write one sentence about it. This supports creativity and ownership.

#### Step 4: Partner Sharing

- Ask students to turn to a partner and complete these sentence starters:

  "One thing I learned about helping people at home is..."

  "Next time, I would try to..."

- This promotes listening, communication, and peer connection.

#### Step 5: Celebrate Growth

- Say:

  "You've done amazing work as Inclusive Home Designers. You thought of ways to help others and made smart choices!"

- Optional: Hand out a small sticker or badge labeled "Smart Home Helper" or "Inclusive Engineer."

# Relation to Standards

## 1. Science Standards (NGSS-aligned)

### Structure and Properties of Matter

- Identify materials in objects, such as wood, metal, or plastic
- Describe how materials' properties (e.g., flexibility, strength) affect their use in a design solution

### Force and Motion

- Understand how push/pull (e.g., from opening smart doors) affects movement

### Engineering Design (ETS1)

- Ask questions, make observations, and gather information about a problem
- Develop a simple sketch or model to show how a device solves a specific need
- Analyze data from tests of objects designed to solve the same problem

## 2. Math Standards (CCSS-aligned)

### Measurement & Data

- Estimate and compare lengths (e.g., designing hallway widths or shelf heights)
- Use tools to measure and reason about object size, height, or distance

### Place Value & Number Comparison

- Write and compare 2- or 3-digit numbers to discuss room sizes, measurements, or model costs

## 3. Technology Standards (ISTE-aligned)

### 1.4 Innovative Designer

- Use a design process to solve open-ended problems with creativity
- Create prototypes using digital tools like PictoBlox

### 1.1 Empowered Learner

- Understand the basics of how AI (speech recognition) can help people

## 4. Social Studies Standards

### Civic Engagement / Social Awareness

- Recognize the roles of community helpers (Assistive Tech Specialists)
- Demonstrate empathy and social responsibility in solving real-world problems
- Community & Inclusivity
- Explore how people adapt spaces to meet different needs (e.g., accessible homes)